

A Padovan-automatic description of a nested recurrence

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Abstract

We study the sequence $a(0) = 0$, $a(1) = 1$ and $a(n) = n - a(n - a(n - 1))$ for $n \geq 2$, listed as A076502 in the On-Line Encyclopedia of Integer Sequences. We identify $a(n)$ as a two-position shift in the greedy Padovan numeration system, with a finite-state correction. The proof constructs an addition automaton from an exact integer-carry invariant and certifies its completeness by finite-language inclusion; a synchronized automaton then verifies the nested recurrence. We establish bounded discrepancy from the line of slope c , where $c^3 - c^2 + 2c - 1 = 0$, and show that the exact set of offsets from $\lfloor cn \rfloor$ is $\{-1, 0, 1, 2\}$. We construct an explicit 26-letter non-erasing morphic presentation of the first-difference word, prove that its least balance constant is 4, and give an effective procedure for enclosing the global discrepancy extrema to arbitrary accuracy. We formalize the recurrence identification, six-decimal discrepancy bound, exact offset set, concrete morphic identity, least balance constant, and an effective extrema algorithm in Lean. Separate exact computations refine the numerical enclosures.

1 Introduction

Consider the nested recurrence

$$a(0) = 0, \quad a(1) = 1, \quad a(n) = n - a(n - a(n - 1)) \quad (n \geq 2). \quad (1.1)$$

Its positive-index terms form sequence A076502 in the OEIS [3]. An original conjectural description placed $a(n)$ within the three offsets 0, 1, 2 of $\lfloor cn \rfloor$, where c is the real root of

$$P(t) = t^3 - t^2 + 2t - 1. \quad (1.2)$$

The example $a(1167) = 664$ and $\lfloor 1167c \rfloor = 665$ rules out that offset set. We establish bounded discrepancy and the asymptotic relation $a(n) \sim cn$, and prove that the exact offset set is $\{-1, 0, 1, 2\}$.

Let $\rho > 1$ be the plastic number, the real root of $\rho^3 = \rho + 1$ (OEIS A060006), and set $c = \rho^{-2}$. Substitution shows $P(c) = 0$; moreover $P'(t) = 3t^2 - 2t + 2 > 0$, so this specifies the same constant as in (1.2). Our numeration system uses

$$U_0, U_1, U_2, U_3, U_4 = 1, 2, 3, 4, 5, \quad U_p = U_{p-2} + U_{p-3} \quad (p \geq 4). \quad (1.3)$$

These are the distinct positive Padovan numbers (OEIS A000931). The recurrence $U_{n+4} = U_{n+2} + U_{n+1}$, valid for all $n \geq 0$, has characteristic polynomial $X(X^3 - X - 1)$: the root 0 accounts for an initial transient. Unlike the canonical $k = 5$ polynomial of [8], $X^5 - X^4 - 1 = (X^2 - X + 1)(X^3 - X - 1)$, it

has no unit-circle roots. Digits are read from most to least significant. Write $\text{rep}(n)$ for the canonical greedy representation of n , with $\text{rep}(0) = \varepsilon$. If its selected positions are J , define

$$\text{shift}_2(n) = \sum_{\substack{p \in J \\ p \geq 2}} U_{p-2}. \quad (1.4)$$

Thus the shift discards the two least significant digits. A deterministic finite automaton with output (DFAO) assigns an output to the state reached after reading a word.

Theorem 1.1. *The recurrence (1.1) defines a unique sequence of nonnegative integers. For the explicit 28-state DFAOs E, D in Appendix Table S2,*

$$a(n) = \text{shift}_2(n) + E(\text{rep}(n)), \quad (1.5)$$

$$a(n+1) - a(n) = D(\text{rep}(n)) \in \{0, 1\} \quad (1.6)$$

for every $n \geq 0$. Its discrepancy satisfies

$$-1.049234 < a(n) - cn < 1.155081 \quad (n \geq 0). \quad (1.7)$$

Consequently $a(n)/n \rightarrow c$, and

$$\{a(n) - \lfloor cn \rfloor : n \geq 1\} = \{-1, 0, 1, 2\}. \quad (1.8)$$

The difference word is morphic, not ultimately periodic, and has least balance constant 4. It admits the explicit 26-letter non-erasing presentation in Table 2. The global infimum and supremum of $a(n) - cn$ are effectively approximable by rational intervals with a certified error tending to zero.

The central step is (1.5). We build an addition automaton for the Padovan system, prove it sound by a carry invariant and complete by a finite inclusion of languages, and verify the recurrence on synchronized representations. Strong induction identifies the resulting function with (1.1). The remaining conclusions follow from a contracting quadratic form, finite weighted-path optimization, and the representation tree.

Nested recurrences with morphic interpretations have been studied systematically by Celaya and Ruskey [1]. Shallit's treatment of the Narayana system [11] is a close methodological precedent: it uses synchronized automata and first-order verification to establish identities for related nested recurrences. Our proof uses this automata method with a directly certified adder for the particular weights (1.3). The passage from abstract-numeration automaticity to morphicity is due to Rigo and Maes [10]; we check its representation-order hypothesis explicitly and also provide the substitution itself. Our contributions are the identification of A076502, its discrepancy bounds, and an explicit morphic presentation.

Recent work of Letouzey, Li, and Steiner [9], and of Letouzey [8], develops the canonical-numeration, shift, and discrepancy approach for generalized Hofstadter functions. The latter work also supplies formalized finite optimization and tail bounds for global discrepancy extrema. Section 4 follows their finite-tail method. Their iterated-composition recurrence differs from (1.1), so identifying A076502 and handling its exceptional initial weight require separate arguments. We specialize the method to the correction automaton E and use a real quadratic form in place of explicit complex-root estimates.

Finite-state normalization is classical in Pisot bases [4] and in linear numeration systems whose characteristic polynomial is the minimal polynomial of a Pisot number [6]; see also [5] on finite beta-expansions and [7] on regular greedy languages. There is an initial-value issue here: the minimal cubic recurrence fails at $p = 3$, since $4 \neq 2 + 1$. Indeed, a strictly increasing integer basis starting with $U_0 = 1$ cannot satisfy $U_3 = U_1 + U_0$, since no integer U_2 lies strictly between U_1 and $U_1 + 1$. The extended annihilator $X(X^5 - X^4 - 1)$ follows by multiplication by $X^2 - X + 1$, but does not remove this exception. We prove greedy regularity and addition directly for these initial values, using a final-digit carry correction to handle the exceptional weight.

2 Greedy representations and certified addition

For a binary word $w = w_k \cdots w_0$, let $\text{val}(w) = \sum_{p=0}^k w_p U_p$. Let G denote the language of greedy representations with arbitrary leading zeros, including the empty word.

Lemma 2.1. *A binary word belongs to G if and only if consecutive positions carrying 1 are at distance at least five, with the sole exception that positions 4 and 0 may both carry 1. Every nonnegative integer has a unique representation after leading zeros are removed. A legal word supported on positions $0, \dots, m$ has value at most $U_{m+1} - 1$.*

Proof. For $p \geq 5$, the recurrence gives

$$U_{p+1} - U_p = U_{p-4}. \quad (2.1)$$

Together with the initial values, this proves that the weights are strictly increasing. If the greedy algorithm selects U_p , its remainder is smaller than $U_{p+1} - U_p$. For $p \geq 5$ the next selected position is therefore at most $p - 5$. At $p = 4$ the remainder is at most 1, allowing position 0; at $p \leq 3$ the remainder is zero.

Conversely, induct on the largest permitted position m . For $m \leq 3$ only singletons can occur. For $m = 4$ the maximum is $U_4 + U_0 = 6 = U_5 - 1$. If $m \geq 5$ is selected, the lower positions are at most $m - 5$, so induction and (2.1) bound the total by

$$U_m + (U_{m-4} - 1) = U_{m+1} - 1.$$

If m is not selected, use the preceding induction case. The largest selected position is thus the first greedy choice, and the remaining positions are greedy by induction. Existence follows because $U_0 = 1$; the deterministic greedy algorithm proves uniqueness. $\square$

The exception is positional: 10001 is legal, but 100010 is not. A partial DFA for G records the number of zeros still required before the next 1, together with a terminal state for the exceptional final 1. Every state of this partial automaton is accepting.

2.1 An exact carry invariant

The weights do not satisfy the homogeneous recurrence at $p = 3$. To account for this, put $W_0 = 2, W_1 = 2, W_2 = 3$ and $W_p = W_{p-2} + W_{p-3}$ for $p \geq 3$. Then $W_p = U_p$ for $p \geq 1$ and $W_0 = U_0 + 1$.

Lemma 2.2. *Read a signed digit word from most to least significant, starting with $r = (0, 0, 0)$ and updating on digit z by*

$$T(r, z) = (r_2 + z, r_0 + r_2, r_1), \quad \Lambda(r) = 2r_0 + 2r_1 + 3r_2. \quad (2.2)$$

After the final digit z_0 , its represented value is

$$\sum_p z_p U_p = \Lambda(r) - z_0. \quad (2.3)$$

The empty word has both sides zero.

Proof. The update multiplies the digit polynomial by X , appends z , and reduces modulo $X^3 - X - 1$. The linear map $X^p \mapsto W_p$ vanishes on multiples of that polynomial, by the homogeneous recurrence. Its value on the remainder is $2r_0 + 2r_1 + 3r_2$. Replacing W by U subtracts exactly z_0 . $\square$

For addition on digit tracks x, y, z , feed the signed digits $x + y - z$ to (2.2). Retain only carries in $[-8, 8]^3$, remember the last signed digit, impose membership in G on all three tracks, and accept when $\Lambda(r)$ equals that last digit. Trim and minimize the resulting partial DFA. It has 411 states; throughout this paper state counts exclude the implicit rejecting sink.

Proposition 2.3 (Certified addition). *Every triple accepted by the resulting relation Add satisfies $x + y = z$ on represented values. Moreover, for every equally long pair of words in G , prefixing three zero columns gives an input pair with an accepted output.*

Proof. Soundness follows immediately from Lemma 2.2 and is independent of the carry cutoff. Completeness is the following finite-language inclusion, checked exhaustively by projection, determinization and product reachability:

$$000(G \times G) \subseteq \exists z \text{Add}(x, y, z). \quad (2.4)$$

Here 000 means three zero columns on both input tracks; the quantified output has the same total length. Every pair of integers has an equally padded input representation of this form. An accepted output exists by (2.4), and its value is the sum by soundness. $\square$

The bound on the carries only restricts the candidate automaton; completeness comes from the inclusion (2.4).

3 Identification with the recurrence

Define

$$b(n) = \text{shift}_2(n) + E(\text{rep}(n)).$$

State 0 of E loops on digit 0, so leading zeros do not change b . To construct its synchronized graph, use a two-digit delay on the input track. It produces the digits of the shifted word, padded with two leading zeros. Apply the carry update to the difference between these delayed digits and the output digits, and run E on the input. The acceptance condition is

$$\Lambda(r) - z_0 + E(q) = 0,$$

with both tracks in G . Every accepted pair (n, v) then satisfies $v = b(n)$. Using the same finite carry cutoff, trimming and minimization give a 72-state relation B .

Proposition 3.1. *The graph relation is total on $000G$, and its represented function satisfies*

$$b(0) = 0, \quad b(1) = 1, \quad 1 \leq b(n) \leq n \quad (n \geq 1). \quad (3.1)$$

Proof. Projection and inclusion verify $000G \subseteq \exists v B(n, v)$. Soundness makes every such output equal to the integer formula defining b , so in particular b is nonnegative. The language of inputs satisfying $\exists v, t [B(n, v) \wedge \text{Add}(v, t, n)]$ contains $000G$. The language satisfying $B(n, 0)$ contains no representation of a positive integer. These two finite checks give the inequalities. The initial values are read directly from E . $\square$

Intersections implement conjunctions of synchronized relations; existential projection removes tracks. To verify the recurrence, define $\text{Good}(n)$ by

$$\begin{aligned} \exists m, p, h, v, k, w, u \ [& m + 1 = n \wedge B(m, p) \wedge h + p = n \\ & \wedge B(h, v) \wedge k + v = n \wedge B(k, w) \\ & \wedge B(n, u) \wedge u + w = n]. \end{aligned} \quad (3.2)$$

Every addition here is implemented with Add . Some intermediate relations are listed in Table 1.

Represented relation	States
$x + y = z$	411
$v = b(n)$	72
$p = b(n - 1), n \geq 1$	79
$v = b(n - b(n - 1))$	144
$w = b(n - b(n - b(n - 1)))$	80
Good(n)	7

Table 1: Selected partial DFAs used in the recurrence verification.

Proposition 3.2. *For every $n \geq 2$,*

$$b(n) + b(n - b(n - b(n - 1))) = n. \quad (3.3)$$

Proof. The exact inclusion check verifies that 000G, restricted to represented values at least 2, is contained in Good. Soundness of each component of (3.2) then gives (3.3) for every such integer. The witnesses are natural numbers. By (3.1), their values are at most n , so their canonical representations fit within that of n ; the three additional leading zeros provide the input padding used in the component checks, so the projections lose no witness. $\square$

Proof of the identification. For $n \geq 2$, (3.1) gives

$$1 \leq n - b(n - 1) < n, \quad 1 \leq n - b(n - b(n - 1)) < n.$$

The first inequality uses $1 \leq b(n - 1) \leq n - 1$; the second applies the same bounds to the first inner index. Starting with $a(0) = b(0)$ and $a(1) = b(1)$, strong induction shows that (1.1) is well-defined and agrees with b at every index. This proves existence, uniqueness, nonnegativity, and (1.5). $\square$

For the difference identity, synchronize the graph at n and $n + 1$ with the output of D and addition. The resulting 12-state input relation expresses $b(n) + D(\text{rep}(n)) = b(n + 1)$; exact inclusion verifies that it contains 000G. This proves (1.6), including $D(\varepsilon) = 1$ at $n = 0$.

4 Discrepancy and effective extrema

Set $f(n) = a(n) - cn$. Define

$$\delta_0 = -c, \quad \delta_1 = -2c, \quad \delta_p = U_{p-2} - cU_p \quad (p \geq 2).$$

By (1.5), for selected positions J ,

$$f(n) = E(\text{rep}(n)) + \sum_{p \in J} \delta_p. \quad (4.1)$$

4.1 A contracting quadratic form

Lemma 4.1. *For $p \geq 3$,*

$$|\delta_p| < \frac{5}{16} \left(\frac{7}{8} \right)^{p-3}. \quad (4.2)$$

Proof. For $i \geq 1$, let $F_i = U_{i+2} + \rho U_{i+1} + \rho^{-1} U_i$. The weight recurrence and $\rho^2 = 1 + \rho^{-1}$ give $F_{i+1} = \rho F_i$. Consequently

$$\delta_{p+2} + \rho \delta_{p+1} + \rho^{-1} \delta_p = 0 \quad (p \geq 3). \quad (4.3)$$

The cutoff $p \geq 3$ ensures that both homogeneous identities used here avoid the anomalous initial weight.

Consider the positive quadratic form

$$\begin{aligned} Q(x, y) &= \rho^{-1}x^2 + \rho xy + y^2 \\ &= \left(y + \frac{\rho x}{2}\right)^2 + \frac{3-\rho}{4\rho}x^2. \end{aligned}$$

Direct expansion gives

$$Q(y, -\rho y - \rho^{-1}x) = \rho^{-1}Q(x, y). \quad (4.4)$$

All constants can be bounded rationally. The signs of $t^3 - t - 1$ at $331/250$ and $53/40$ give $331/250 < \rho < 53/40$. Hence $\rho > 64/49$, $\rho^{-1} < 19/25$, $\rho > 33/25$, and $(3 - \rho)/(4\rho) > 5/16$. Using $\delta_3 = 2 - 4c$ and $\delta_4 = 3 - 5c$, these brackets imply

$$-\frac{141}{500} < \delta_3 < -\frac{139}{500}, \quad \frac{147}{1000} < \delta_4 < \frac{153}{1000}.$$

Keeping the sign of the mixed term,

$$\begin{aligned} Q(\delta_3, \delta_4) &< \frac{19}{25} \left(\frac{141}{500}\right)^2 + \left(\frac{153}{1000}\right)^2 - \frac{33}{25} \frac{139}{500} \frac{147}{1000} \\ &= \frac{747603}{25000000} < \frac{3}{100}. \end{aligned}$$

Iterating (4.4) along (4.3) gives

$$|\delta_p|^2 < \frac{12}{125} \left(\frac{49}{64}\right)^{p-3} < \frac{25}{256} \left(\frac{49}{64}\right)^{p-3},$$

which proves (4.2). $\square$

For $K \geq 5$, selected positions at least K are spaced by at least five. The absolute contribution of this tail is therefore strictly smaller than

$$R_K = \frac{(5/16)(7/8)^{K-3}}{1 - (7/8)^5}. \quad (4.5)$$

The finitely many lower positions and the bounded output E already imply that f is bounded. To obtain useful constants we optimize the lower digits jointly with the automaton state.

4.2 Suffix optimization and realizable witnesses

Following the finite-tail method of [8], we retain the correction state and reaching prefixes so that extremal paths represent actual integers. Let $\alpha < c < \beta$ be a rational bracket. After an arbitrary high prefix, there are 28 reachable pairs consisting of an E -state and a G -state. For each such pair fix a finite reaching prefix. A breadth-first search finds prefixes of length at most $H = 13$.

For each legal path of length K from these states, its contribution from the last K digits, including the final output of E , has the form $A - cB$, where $A \in \mathbb{Z}$ and $0 \leq B < U_K$. Dynamic programming on the finite state set computes the exact rational extrema

$$L_K = \min(A - \beta B), \quad V_K = \max(A - \alpha B), \quad (4.6)$$

retaining one path attaining each endpoint. By (4.5),

$$L_K - R_K < f(n) < V_K + R_K \quad (n \geq 0). \quad (4.7)$$

Short representations are padded with zeros; this changes neither legality nor the output of E .

Proposition 4.2 (Effective global extrema). *Let $m = \inf_{n \geq 0} f(n)$ and $M = \sup_{n \geq 0} f(n)$. The suffix construction provides rational enclosures of both numbers with width at most*

$$2R_K + (\beta - \alpha)(U_K + U_{K+H}), \quad H = 13. \quad (4.8)$$

Choosing $\beta - \alpha = 2^{-4K-64}$ gives widths tending to zero.

Proof. Attach the recorded prefixes to extremizing suffixes and bound the omitted tail by R_K . The full argument and witness enclosures are given in Appendix S1. $\square$

4.3 Numerical certificates and the corrected offsets

Exact rational optimization at cutoff $K = 200$ gives the outward-rounded enclosures below; the cutoff table and witnesses are in Appendix S1. The weaker six-decimal bound (1.7) is also checked in Lean using a separate $K = 105$ certificate. The refined enclosures are

$$-1.049233035332 \leq m \leq -1.049233035329, \quad (4.9)$$

$$1.155080224901 \leq M \leq 1.155080224905. \quad (4.10)$$

We do not know whether these extrema are attained.

Put $k = a(n) - \lfloor cn \rfloor$. Since $k = f(n) + \{cn\}$, (1.7) gives $-2 < k < 3$. The integer k belongs to $\{-1, 0, 1, 2\}$. All four values occur:

n	$a(n)$	$\lfloor cn \rfloor$	Offset
1167	664	665	-1
2	1	1	0
1	1	0	1
93	54	52	2

The floor values are certified by the signs at consecutive integers of $k^3 - k^2n + 2kn^2 - n^3 = n^3P(k/n)$. The sequence values at these small indices can be computed directly from (1.1). Dividing (1.7) by n proves $a(n)/n \rightarrow c$.

5 A morphic description of the difference word

Let $d(n) = a(n+1) - a(n)$ for $n \geq 0$. Its first symbol is $d(0) = 1$; thus the positive-index difference word is obtained by deleting this initial symbol. Define the canonical language

$$L = \{\varepsilon\} \cup (G \cap 1\{0, 1\}^*), \quad 0 < 1.$$

Lemma 5.1. *Genealogical order on L , namely length followed by lexicographic order, coincides with increasing represented value.*

Proof. A canonical word of length $p+1$ has value at least U_p , whereas a shorter word has value at most $U_p - 1$ by Lemma 2.1. For equal lengths, consider the largest position p where two words differ. The lexicographically larger word contributes U_p there; the entire lower suffix of the smaller word contributes at most $U_p - 1$. Their higher parts agree, proving strict value order. Greedy existence and uniqueness complete the claim. $\square$

The language L is infinite and regular and contains ε . Therefore $(L, \{0, 1\}, <)$ is an abstract numeration system enumerating precisely $\text{rep}(0), \text{rep}(1), \dots$. The verified D -DFAO makes d automatic in this system. The Rigo–Maes theorem [10], explicitly restated in the one-dimensional case of [2,

Theorem 24], implies that d is morphic. The canonical language L gives each integer a unique representation, as required for this enumeration.

Our Lean proof of the concrete presentation uses the following direct argument. For each fixed length m , the legal padded words enumerate exactly the integers below U_m . Their lexicographic rank equals their represented value: whenever a digit 1 is legal, the preceding digit-0 subtree has size U_j , where j is the remaining suffix length. Induction on the input word proves that the corresponding iterate of the state morphism has the correct D -state at every index. Fixing the length gives each integer in this range a unique padded representation.

5.1 A non-erasing presentation

We give an explicit presentation. Form the product of D with the partial automaton for G , starting after the first digit 1. Exactly 27 pairs are reachable, with distinct D -states $q_1, \dots, q_{27}$. Only q_6 and q_{21} have no legal children. Every other state has a nonterminal child on digit 0. These statements are checked by exhaustive traversal of the finite product; the script also verifies that D is defined on every legal continuation.

Introduce a marker S with output 1, corresponding to ε . Define a morphism μ by

$$\mu(S) = Sq_1, \quad \mu(q) = \text{the legal children of } q \text{ in digit order } 0, 1.$$

In particular $\mu(q_6) = \mu(q_{21}) = \varepsilon$. Its fixed point

$$x = \mu^\omega(S) = Sq_1\mu(q_1)\mu^2(q_1)\cdots$$

enumerates the representation tree level by level. Every level contains $10 \cdots 0$, so the word is infinite. Let τ output 1 on S and the D -output on each q_i . By Lemma 5.1 and (1.6), $\tau(x) = d(0)d(1)\cdots$.

Proposition 5.2. *Let $I = \{S, q_1, \dots, q_{27}\} \setminus \{q_6, q_{21}\}$. There are non-erasing morphisms $\nu : I^* \rightarrow I^*$ and $\eta : I^* \rightarrow \{0, 1\}^*$, given in Table 2, such that*

$$d(0)d(1)d(2)\cdots = \eta(\nu^\omega(S)). \quad (5.1)$$

Proof. Let π delete q_6, q_{21} and fix every other letter. On I , put $\nu = \pi\mu$, $h = \mu$, and $\eta = \tau h$. Since every nonterminal state has a nonterminal child, both ν and η are non-erasing. The relation $\pi\mu = \nu\pi$ gives $y = \pi(x) = \nu^\omega(S)$. Because μ annihilates the deleted letters,

$$x = \mu(x) = \mu(\pi(x)) = h(y).$$

Applying τ proves (5.1). □

The output morphism η has images of length one or two; it is not a letter coding on this same alphabet. We do not know whether this presentation is minimal, nor how it relates to the six-letter substitution found experimentally by the first author.

Corollary 5.3. *The frequency of 1 in d is c , the word is not ultimately periodic, and its least balance constant is 4.*

Proof. There are $a(N)$ ones in the first N symbols, so their frequency tends to c . The rational-root test shows that P has no rational root; an ultimately periodic binary word would have rational letter frequency. For a length- ℓ factor beginning at i , the number of ones is

$$a(i + \ell) - a(i) = c\ell + f(i + \ell) - f(i).$$

The bounds in (1.7) have width 2.204315. Counts of ones in two equal-length factors thus differ by less than 4.408630, hence by at most 4 since the difference is integral. The same holds for zeros. For sharpness, direct evaluation gives $a(135) = 76$, $a(207) = 119$, $a(214) = 123$, and $a(286) = 162$. The length-72 factors starting at 135 and 214 therefore contain 43 and 39 ones, respectively. Their difference is 4, ruling out every smaller balance constant. These evaluations are also kernel checked. □

q	$\nu(q)$	$\eta(q)$	q	$\nu(q)$	$\eta(q)$
S	$S q_1$	10	q_{14}	q_{17}	0
q_1	q_2	1	q_{15}	q_{18}	1
q_2	q_3	0	q_{16}	q_{19}	0
q_3	q_4	1	q_{17}	q_{20}	01
q_4	q_5	10	q_{18}	q_{22}	0
q_5	$q_7 q_8$	01	q_{19}	q_{23}	11
q_7	$q_9 q_{10}$	11	q_{20}	$q_{24} q_1$	10
q_8	q_{11}	0	q_{22}	q_9	11
q_9	$q_5 q_{12}$	10	q_{23}	$q_{25} q_{10}$	01
q_{10}	q_{13}	0	q_{24}	$q_{26} q_1$	00
q_{11}	q_{14}	1	q_{25}	$q_{20} q_{27}$	01
q_{12}	q_{15}	1	q_{26}	$q_{23} q_8$	11
q_{13}	q_{16}	1	q_{27}	q_{15}	1

Table 2: The non-erasing substitution and output morphism in [Proposition 5.2](#). The initial letter is S .

6 Verification and further questions

The finite-state assertions are proved by exhaustive language inclusion, using integer transition tables. Two separately written Python implementations reproduce the recurrence checks, and Walnut independently reconstructs six quantified relations. The accompanying Lean development checks the entire identification chain, the six-decimal global bound, the exact offset set, the concrete morphic identity, the frequency, nonperiodicity, and the least balance constant 4. Its final axiom closures contain only `propext`, `Classical.choice`, and `Quot.sound`.

Lean also checks a computable extrema algorithm: exhaustive search over $n < U_{K+16}$ with rational root brackets and a proved geometric error. The faster suffix optimizer used for (4.9)–(4.10) is a separate exact computation; those narrower decimal enclosures are not Lean theorems. Appendix S2 records verification scope and reproducible build instructions.

Equality with the original six-letter candidate remains open, as do minimality of the morphic alphabet, attainment of the global extrema, and their relation to asymptotic extrema. The dynamical description of the difference word, for instance as a coding of a toral rotation, remains open.

Contributions and use of AI systems

B.C. identified the Padovan description of the sequence as a two-position shift with a finite-state correction, supplied the correction automata, and wrote an independent verification of the automata proof. W.Z. and H.M. found the counterexample to the original offset conjecture, built the certified adder, the identification proof and the discrepancy bounds, and developed the Lean and Walnut verifications.

The first author used Claude (Anthropic) as an experimental laboratory for computations, including verification code, and for cross-checks against the literature. H.M. and W.Z. used OpenAI models, primarily GPT-5.6 and GPT-6 Astra, through Codex to assist with counterexample searches, exploration of proof strategies, automata construction, and the writing, debugging, and review of Python code, Lean proofs, and Walnut specifications. These tools also assisted with cross-checking implementations and computational results, comparing the arguments with the literature, and drafting and revising the manuscript.

Model-generated suggestions and code were subjected to the formal and computational checks described in this paper. The Lean kernel checked the formalized theorem statements and proofs; Appendix S2 records their scope and axiom closures. All authors retain responsibility for the mathematical claims, verification code, and final text.

S1 Effective extrema: proof and numerical certificates

Write $f(n) = a(n) - cn$, $m = \inf_{n \geq 0} f(n)$, and $M = \sup_{n \geq 0} f(n)$. For $K \geq 5$, the sparse-tail bound is

$$R_K = \frac{(5/16)(7/8)^{K-3}}{1 - (7/8)^5}.$$

Let $\alpha < c < \beta$ be a rational bracket. A legal suffix of length K , including the correction output, contributes $A - cB$, where $A \in \mathbb{Z}$ and $0 \leq B < U_K$. Optimize over suffix paths from the 28 reachable correction/greedy state pairs:

$$L_K = \min(A - \beta B), \quad V_K = \max(A - \alpha B).$$

Each pair has a recorded reaching prefix of length at most $H = 13$. The global bound is $L_K - R_K < f(n) < V_K + R_K$.

S1.1 Proof of the effective-extrema proposition

Join a minimizing suffix path to its recorded reaching prefix, producing a legal word for an integer n_- . Likewise obtain n_+ from a maximizing path. Each concatenation is legal because its prefix reaches the initial state of the chosen suffix. Since their lengths are at most $K + H$, both integers are smaller than U_{K+H} by the greedy-language bound in the main text.

Use the representation formula and root bracket to enclose their discrepancies. If u_- is the upper endpoint for $f(n_-)$ and ℓ_+ is the lower endpoint for $f(n_+)$, then

$$L_K - R_K \leq m \leq u_-, \quad \ell_+ \leq M \leq V_K + R_K. \quad (\text{S1.1})$$

The chosen minimizing suffix has true value at most $L_K + (\beta - \alpha)U_K$. Its high-prefix contribution is less than R_K in absolute value, and enclosing its full discrepancy adds at most $(\beta - \alpha)U_{K+H}$. Thus each enclosure has width at most

$$2R_K + (\beta - \alpha)(U_K + U_{K+H}). \quad (\text{S1.2})$$

The maximum is symmetric. Exact bisection of $P(t) = t^3 - t^2 + 2t - 1$ on $[0, 1]$ supplies a bracket of width 2^{-4K-64} . Since $U_j < 2^{j+1}$, the second term tends to zero; R_K tends to zero geometrically as well. This proves the proposition.

S1.2 Exact computation and witnesses

Table S1 gives rational enclosures of the global extrema with outward-rounded decimal endpoints.

K	Lower endpoint	Upper endpoint	Extrema width
20	-1.086486290247	1.198282591609	0.079247100790
70	-1.049288300721	1.155139949837	0.000096687582
120	-1.049233123804	1.155080305728	0.000000115468
200	-1.049233035332	1.155080224905	0.000000000003

Table S1: Global discrepancy bounds. The last column bounds the larger exact extrema-enclosure width, rounded upwards.

For $K = 200$, the root bracket uses 864 exact bisection steps. Representative witnesses are

$$\begin{aligned} n_- &= 27957586325095713833899543, & a(n_-) &= 15931359127095736187627998, \\ n_+ &= 9231177514959072345795515, & a(n_+) &= 5260296881378963990671881. \end{aligned}$$

The representation formula evaluates these large values directly from their greedy representations. Exact discrepancy brackets and greedy words are recorded in `develop/automata-results/extrema.json`. For suffix lengths 5, 8, 12, exhaustive enumeration agrees with the dynamic program. All eight stored witnesses at four cutoffs are accepted by the synchronized graph, and the two small witnesses also agree with direct recurrence evaluation.

S2 Verification scope and reproduction

S2.1 Finite-state and independent checks

Products implement intersection, subset construction implements existential projection, and inclusion is decided by exhaustive reachability against the complement of the target. Undefined transitions reject. Each inclusion either returns a counterexample or exhausts the finite search space.

Two separately written Python implementations reproduce the principal recurrence checks. The second imports the literal correction table, but none of the first implementation's automata or transformation code. At carry cutoffs 8 and 10 it passes adder totality, graph totality, the recurrence and value bounds. The morphic presentation and representation order were also compared with the literal recurrence through 100,000 terms; this is a diagnostic, not an all-index proof.

Walnut 7.1.0 imports the base adder, candidate graph, greedy domains, constants and difference relation, then reconstructs all intermediate quantified relations. Adder totality, graph totality, value bounds, positivity, the recurrence and the difference identity all return TRUE. Integer interpretation still rests on the carry proof.

S2.2 Lean scope

The Lean development checks canonical greedy representations, recognition, carry and delay semantics, finite-language inclusion and projection, totality, and identification with the literal nested recurrence. It also proves the exact four offsets, slope limit, concrete 26-letter morphic identity, irrational frequency, nonperiodicity, and exact balance constant 4. In `develop/lean/BalanceSharpness.lean`, the two length-72 factors have 43 and 39 ones. The theorem `exact_balance` states that a natural number C is a uniform balance bound if and only if $4 \leq C$.

A real quadratic contraction and sparse tail supply the analytic bound. The $K = 105$ Bellman certificate with a 160-bit dyadic root bracket proves $-1.049234 < f(n) < 1.155081$ in the kernel. The formal extrema algorithm searches $n < U_{K+16}$ and uses a bracket of width 2^{-4K-64} . A prefix-compression lemma bounds the discrepancy outside this window, proving rational enclosures and convergence. This exhaustive algorithm establishes computability; it is distinct from the faster suffix optimizer used for the narrow $K = 200$ decimals.

Final theorem axiom closures contain only `propext`, `Classical.choice`, and `Quot.sound`. No admitted proofs, extra axioms, or native decision axioms are used. Kernel reduction checks the generated finite certificates. The narrow $K = 200$ decimals and the original six-letter identity are not asserted as Lean theorems.

S2.3 Reproduction commands

The reproducibility package is available at <https://github.com/the-omega-institute/a076502-padovan>. Version v1.0.1 is archived at <https://doi.org/10.5281/zenodo.22979217>.

From the package's `develop/` directory, standard-library Python 3 reconstructs the finite-state proof and certificates:

```
python3 automata_proof.py --bound 8
python3 discrepancy_certificate.py
python3 morphic_certificate.py
python3 extrema_certificate.py
python3 paper_checks.py
```

The original literal tables and independent verifier are retained, unmodified, in `correspondence/attachments/`.

Reports in `develop/automata-results/` include `report.json`, `morphic-presentation.json`, `extrema.json`, `paper-checks.json`, and `walnut.json`. The optional million-term diagnostic is `independent_checks.py`. Walnut can be replayed using `walnut_check.py` with Java and the pinned jar described in `develop/walnut/README.md`.

The Lean toolchain is 4.33.0, with Mathlib pinned to

```
db584cd6d46c92f209a44c0f1c829460d327499d.
```

With Elan, Git, Python 3 and network access available, run from the extracted research package root:

```
python3 develop/clean_ci.py
```

This command rebuilds the Lean development in a fresh Lake project with the pinned dependencies and records source hashes and final axiom closures in `develop/ci-results/`. The README of the package gives the details.

S3 The finite-state tables

Table S2 specifies both input transition functions and outputs. The initial state is 0 for both automata. A dash denotes an undefined transition. The transition functions are not identical: state 22 has different digit-1 successors in E and D . Representations are always restricted to the greedy language.

State	E			D		
	0	1	out	0	1	out
0	0	1	0	0	1	1
1	2	–	1	2	–	0
2	3	–	1	3	–	1
3	4	–	1	4	–	0
4	5	6	0	5	6	1
5	7	8	0	7	8	1
6	–	–	1	–	–	0
7	9	10	0	9	10	0
8	11	–	0	11	–	1
9	5	12	0	5	12	1
10	13	–	1	13	–	1
11	14	–	2	14	–	0
12	15	–	1	15	–	0
13	16	–	1	16	–	0
14	17	–	0	17	–	1
15	18	–	1	18	–	1
16	19	–	1	19	–	1
17	20	21	1	20	21	0
18	22	–	1	22	–	1
19	23	21	1	23	21	0
20	24	1	0	24	1	0
21	–	–	0	–	–	1
22	9	6	0	9	21	0
23	25	10	–1	25	10	1
24	26	1	0	26	1	1
25	20	27	1	20	27	0
26	23	8	1	23	8	0
27	15	–	0	15	–	1

Table S2: The E - and D -DFAOs used in the main text. Columns 0 and 1 give successor states; “out” gives the output.

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